Module Project 1: Statistical Analysis Portfolio

A Statistical Exploration of Employee Attrition and Workplace Factors at IBM

Introduction:

This project applies statistical concepts to the IBM HR Employee Attrition dataset, which contains demographic, organizational, and compensation data. By deploying methods from Modules 0–2, we explore descriptive statistics of both categorical and numerical variables, apply probability theory to study relationships between variables such as attrition and overtime, and analyze random variables by constructing probability distributions. The aim is to demonstrate not only the calculations but also meaningful interpretations that connect statistical outcomes to workforce insights.

Module 0: Descriptive Statistics

1. Dataset Overview

The dataset analyzed in this project is the IBM HR Employee Attrition dataset, which includes **1,470 employees** and **35 variables** covering demographics, compensation, work history, and attrition status. A complete list of variables is included in the appendix; for focused statistical analysis and relevance to business questions, we select a subset of variables described below.

2. Variable Selection and Rationale

We chose a mix of categorical and numeric variables since they reflect both individual employee profiles and broader organizational factors that are often tied to attrition; other variables are available for future modules and deeper analysis.

Table 1: Variable Selection and Types

Variable	Type	Rationale
Attrition	Categorical	Core outcome variable for HR/business interest (whether the employee leave or not)
Department	Categorical	Reflects organizational unit, possible driver of attrition
OverTime	Categorical	Indicates workload/stress, potential attrition factor
Age	Numeric	Key demographic context
MonthlyIncome	Numeric	Compensation, linked to satisfaction and retention
YearsAtCompany	Numeric	Employee tenure, crucial for retention analysis

These variables above give a balanced perspective: they allow us to analyze not only who leaves, but also how compensation, demographics, and organizational context contribute to retention. This makes the analysis meaningful not just for the dataset, but also for real HR challenges organizations face today, like managing turnover, ensuring fair pay, and supporting employees at different career stages.

3. Description of Data

Table 2: Summary Table – Categorical Variables

Attrition	No	Yes
Count	1233	237
Rate	0.8387755	0.1612245

Department	Human Resources	Research & Development	Sales
Count	63	961	446
Rate	0.04285714	0.65374150	0.30340136

OverTime	No	Yes
Count	1054	416
Rate	0.7170068	0.2829932

Interpretation: The categorical variables reveal key patterns in IBM’s workforce composition and potential retention challenges. Out of 1470 total employees, 83.88% (1233) stayed with the company while 16.12% (237) left. This means roughly one in six employees experienced attrition, which is a moderate turnover rate that still represents significant replacement costs for the organization.

Looking at department distribution, R&D dominates with 961 employees (65.37%), followed by Sales with 446 (30.34%), and HR with just 63 (4.29%). This heavy concentration in R&D makes sense for IBM as a technology company where innovation drives competitive advantage. The substantial Sales presence shows the importance of client relationships, while HR's small size is typical for support functions.

The overtime data reveals a concerning pattern: 28.30% of employees (416) regularly work overtime while 71.70% (1054) maintain standard hours. Nearly three in ten employees working beyond regular hours suggests potential workload imbalances or understaffing that could contribute to burnout and eventual attrition.

Table 3: Summary Table – Numerical Variables

Variable	Min	Q1	Median	Mean	Q3	Max
Age	18.00	30.00	36.00	36.92	43.00	60.00
MonthlyIncome	1009	2911	4919	6503	8379	19999
YearsAtCompany	0.000	3.000	5.000	7.008	9.000	40.000

Interpretation:

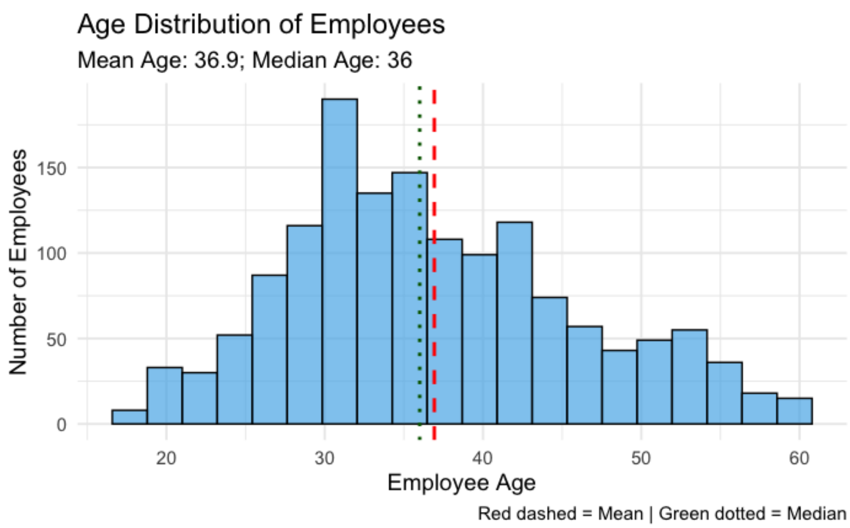
The numerical variables provide insight into IBM's workforce demographics and compensation structure. The age distribution shows a median of 36 years and a mean of 36.92 years - these nearly identical values indicate a symmetric age distribution with employees ranging from 18 to 60. The workforce clusters around experienced mid-career professionals, with the first quartile at 30 and third quartile at 43, suggesting IBM attracts and retains employees in their prime working years.

Monthly income reveals significant pay disparities across the organization. The median income of $4,919 sits well below the mean of $6,503, indicating a right-skewed distribution where most employees earn less than the average. This gap between median and mean suggests a subset of high earners pulling the average up. With the minimum at $1,009 and maximum at $19,999, there's nearly a 20-fold difference between lowest and highest paid employees.

Years at company shows interesting retention patterns with a mean of 7 years but high variation (Q1=3, Q3=9). The minimum of 0 and maximum of 40 years indicates IBM has both brand-new hires and extremely long-tenured employees. The median of 5 years suggests that if employees make it past the initial adjustment period, many tend to stay for extended periods.

4. Visualizations

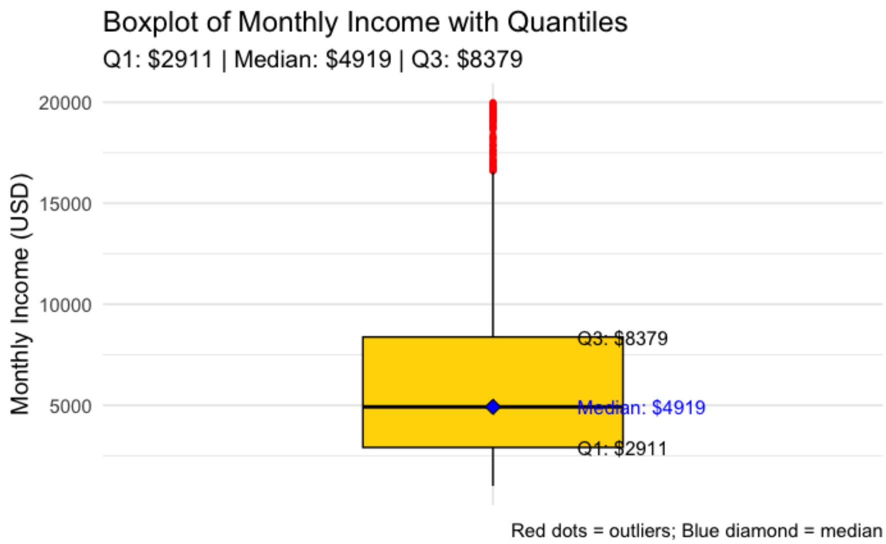
Figure 1: Histogram of employee ages, with mean (red dashed) and median (green dotted) lines.



Interpretation: The age distribution is centered around the mid-30s, with most employees between 30–40 years old. The mean (36.9) and median (36) are nearly identical, indicating a fairly symmetric distribution. This suggests a predominantly mid-career workforce, balanced between younger entrants and more experienced employees.

From a workforce perspective, this indicates that the company has a mid-career employee base, with a good balance of younger employees entering the workforce and more experienced employees approaching later stages of their career. This age structure is typical of many organizations, providing both growth potential from younger workers and stability from experienced staff.

Figure 2: Boxplot of monthly income, showing median, quartiles, and outliers.



Interpretation: The age distribution shows that the mean (36.9) and median (36) are nearly identical, differing by less than one year. This close alignment indicates a fairly symmetric age structure, without strong skewness. In practical terms, IBM's workforce spans a wide range of career stages, from early-career to experienced employees, without clustering in a narrow age band.

The monthly income boxplot paints a different picture. Here, the substantial gap between the median ($4,919) and the mean ($6,503) confirms a right-skewed income distribution. The numerous outliers above the upper whisker represent high earners—likely executives or senior technical staff—whose salaries significantly pull the average upward. Despite this skewness, the majority of employees earn between $3,000 and $8,000, making this the central range of compensation.

Module 1: Probability Analysis

We selected Attrition and OverTime as our primary variables for probability analysis because overtime represents one of the most direct and measurable indicators of work-life balance issues. Unlike subjective measures like job satisfaction, overtime is objective - either an employee regularly works beyond standard hours, or they don't. The relationship between excessive work hours and employee burnout is well-documented across industries, making this a logical starting point for understanding attrition patterns. If overtime significantly increases the probability of leaving, this suggests a relatively straightforward intervention opportunity for IBM - better workload management could directly impact retention. Since leaving the company and being overworked through overtime is the CLEAR correlation that most people can think of, we can analysis the other parts of it later, but this is the most important correlation.

Selected Variables: Attrition and OverTime

<u>Choice of Methods:</u>

1. Joint Probability:

The **joint probability** of events A and B happening together is:

$$P(A \cap B) = P(A \; and \; B)$$

Or, for discrete variables, if you have a contingency table:

$$P(A = a, B = b) = \frac{Number \; of \; cases \; where \; A = a \; and \; B = b}{Total \; Number \; of \; Cases}$$

Table 4: Contingency Table - Attrition vs OverTime

	OverTime = No	OverTime = Yes	Total
Attrition = No	944	289	1233
Attrition = Yes	110	127	237
Total	1054	416	1470

Interpretation: The contingency table lays out the raw relationship between overtime and attrition. Out of 1054 employees who don't work overtime, 944 stayed and 110 left. Among the 416 who do work overtime, 289 stayed but 127 left. Right away, we can see that the overtime group has a higher proportion leaving (127 out of 416) compared to the regular hours group (110 out of 1054). This suggests overtime and attrition are indeed connected, though we need probability calculations to quantify exactly how strong this relationship is.

Table 5: Joint Probability Table - Attrition vs OverTime

	OverTime = No	OverTime = Yes	Total
Attrition = No	0.6422	0.1966	0.8388
Attrition = Yes	0.0748	0.0864	0.1612
Total	0.7170	0.2830	1.0

Interpretation: The joint probability table shows what percentage of all employees fall into each combination. The key finding is that 8.64% of all employees both worked overtime AND left the company, while only 7.48% worked regular hours and left. Meanwhile, 64.22% worked regular hours and stayed - the largest group by far. These joint probabilities start revealing the pattern that overtime workers are overrepresented among those who leave. While 8.64% might seem small, remember that overtime workers only make up 28.3% of the workforce, so having them represent over half of those who left (127 out of 237) is quite significant.

2. Conditional Probability

The **conditional probability** of event A given B is:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Table 6: Conditional Probability - P (Attrition | OverTime)

	OverTime = No	OverTime = Yes
Attrition = No	0.8956	0.6947
Attrition = Yes	0.1044	0.3053

Interpretation: Now we see the real impact - employees who work overtime have a 30.53% chance of leaving, compared to just 10.44% for those who don't work overtime. This means overtime nearly triples the attrition risk. One in three overtime workers will leave the company, versus only one in ten regular-hours workers. This stark difference suggests overtime is indeed a major factor in employee departures, validating our hypothesis about burnout being a key driver of attrition.

Table 7: Conditional Probability - P (OverTime | Attrition)

	Attrition = No	Attrition = Yes
OverTime = No	0.7656	0.4641
OverTime = Yes	0.2344	0.5359

Interpretation: Flipping the perspective reveals another important insight - among employees who left, 53.59% had been working overtime, even though overtime workers only represent 28.3% of the total workforce. This means overtime workers are dramatically overrepresented in the attrition group. Conversely, among those who stayed, only 23.44% worked overtime. This reverse view reinforces that overtime is a strong predictor of departure.

3. Bayes' Theorem:

   Bayes' Theorem relates conditional probabilities:

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

Where:

- $P(A \cap B)$: Probability that both A and B occur
- $P(A \mid B)$: Probability that A occurs given B occurs
- $P(B \mid A)$: Probability that B occurs given A occurs
- $P(A), P(B)$: Marginal probabilities of A and B

We want to compute:

$P(Attrition = Yes \mid OverTime = Yes)$ by using Bayes' theorem

where:

- A = Attrition = Yes
- B = OverTime = Yes

Then,

$$P(\,Attrition = Yes \mid OverTime = Yes\,) =$$

$$\frac{P(\,OverTime = Yes \mid Attrition = Yes\,) \times P(Attrition = Yes)}{P(OverTime = Yes)}$$

From the data and code output:

$$P(\,OverTime = Yes \mid Attrition = Yes\,) = \frac{127}{237} \approx 0.5359$$

$$P(Attrition = Yes) = \frac{237}{1470} \approx 0.1612$$

$$P(OverTime = Yes) = \frac{416}{1470} \approx 0.2830$$

Plug into Bayes' theorem:

$$P(\,Attrition = Yes \mid OverTime = Yes\,) = \frac{0.5359 \times 0.1612}{0.2830} \approx 0.305$$

Therefore, Probability of Attrition is Yes given OverTime is Yes:

$$P(\,Attrition = Yes \mid OverTime = Yes\,) \approx 0.305\ (30.5\%)$$

Interpretation: Using Bayes' theorem allows us to mathematically connect these different probability perspectives. We calculated that P(Attrition|OverTime) = 30.5%, confirming our conditional probability finding. The beauty of Bayes' theorem is it shows how we can derive this result from the reverse conditional probability (53.59% of leavers worked overtime) combined with the base rates. This 30.5% probability means IBM should flag any employee working consistent overtime as high risk for departure - nearly one in three will leave if this pattern continues.

Having established that overtime significantly impacts attrition risk, we should examine whether department affiliation shows similar patterns. Different departments face unique stressors and work cultures - R&D deals with technical complexity and project deadlines, Sales faces rejection and travel demand, while HR manages interpersonal conflicts and organizational changes. By applying the same probability analysis to departments, we can identify whether certain organizational units are bleeding talent more than others, which would suggest department-specific retention strategies are needed rather than company-wide policies. Let's see if the department someone works in predicts their likelihood of leaving as strongly as overtime does.

Selected Variables: Attrition vs Department

Choice of Method:

1. Joint Probability

Table 8: Contingency Table - Attrition vs Department

	Human Resources	Research & Development	Sales	Total
Attrition = No	51	828	354	1233
Attrition = Yes	12	133	92	237
Total	63	961	446	1470

Interpretation: The department contingency table reveals the raw numbers behind departmental attrition. R&D lost 133 employees, Sales lost 92, and HR lost 12. However, these raw numbers can be misleading since the departments are different sizes. We need to calculate rates to make fair comparisons in the future.

Table 9: Joint Probability Table - Attrition vs Department

	Human Resources	Research & Development	Sales	**Total**
Attrition = No	0.0347	0.5633	0.2408	0.8388
Attrition = Yes	0.0082	0.0905	0.0626	0.1612
Total	0.0429	0.6537	0.3034	1.0

Interpretation: The joint probabilities show that 9.05% of all employees were in R&D and left, 6.26% were in Sales and left, while only 0.82% were in HR and left. But again, these percentages are influenced by department size. R&D's higher percentage partly reflects that it contains 65% of all employees. The more telling analysis comes from conditional probabilities.

2. Conditional Probability

Table 10: Conditional Probability - P (Attrition | Department)

Department	Human Resources	Research & Development	Sales	Total
Attrition = No	0.8095	0.8616	0.7937	1.00
Attrition = Yes	0.1905	0.1384	0.2063	1.00

Interpretation: The conditional probabilities reveal the true departmental attrition rates. Sales has the highest at 20.63% - one in five Sales employees leave. HR follows at 19.05%, while R&D has the lowest at 13.84%. This contradicts any assumption that technical complexity drives people away - actually, R&D retains people better than Sales. The high Sales attrition likely reflects the pressure-cooker nature of sales roles, with quotas, rejections, and travel stress. R&D's lower rate suggests technical employees find their work engaging enough to offset other challenges.

Table 11: Conditional Probability - P (Department | Attrition)

Attrition	Human Resources	Research & Development	Sales	Total
Attrition = No	0.0414	0.6715	0.2871	1.00
Attrition = Yes	0.0506	0.5612	0.3882	1.00

Interpretation: Among employees who left, 56.12% came from R&D, 38.82% from Sales, and 5.06% from HR. While R&D contributes the most leavers in absolute terms, this is proportionally less than their 65% share of the workforce. Sales is overrepresented among leavers (38.82% of departures vs 30.34% of workforce), confirming it's a higher-risk department.

3. Bayes' Theorem: P (Attrition=Yes | Department=Sales)

By Bayes' Theorem,

$$P(\,Attrition = Yes \mid Department = Sales\,)$$

$$= \frac{P(Department = Sales \mid Attrition = Yes) \times P(Attrition = Yes)}{P(Department = Sales)}$$

Plug in data and values from code outputs:

- $P(Department = Sales \mid Attrition = Yes) = 0.3882$
- $P(Attrition = Yes) = \frac{237}{1470} = 0.1612$
- $P(Department = Sales) = \frac{446}{1470} = 0.3034$

Then,

$$P(\,Attrition = Yes \mid Department = Sales\,) = \frac{0.3882 \times 0.1612}{0.3034} \approx 0.2063$$

Interpretation: Using Bayes' theorem, we calculated that Sales employees have a 20.63% probability of leaving. This calculation demonstrates how we can derive departmental attrition rates from the reverse conditional probabilities. With one in five Sales employees departing, IBM should consider department-specific retention strategies, perhaps addressing the unique stressors in sales roles like travel demands and performance pressure.

Module 2: Random Variables, PMF, Expectation, Variance, and CDF

1. Discrete Random Variable: YearsAtCompany

1) Probability Mass Function (PMF)

For a discrete random variable X, the probability mass function is:

$$P(X = x) = p(x)$$

where:

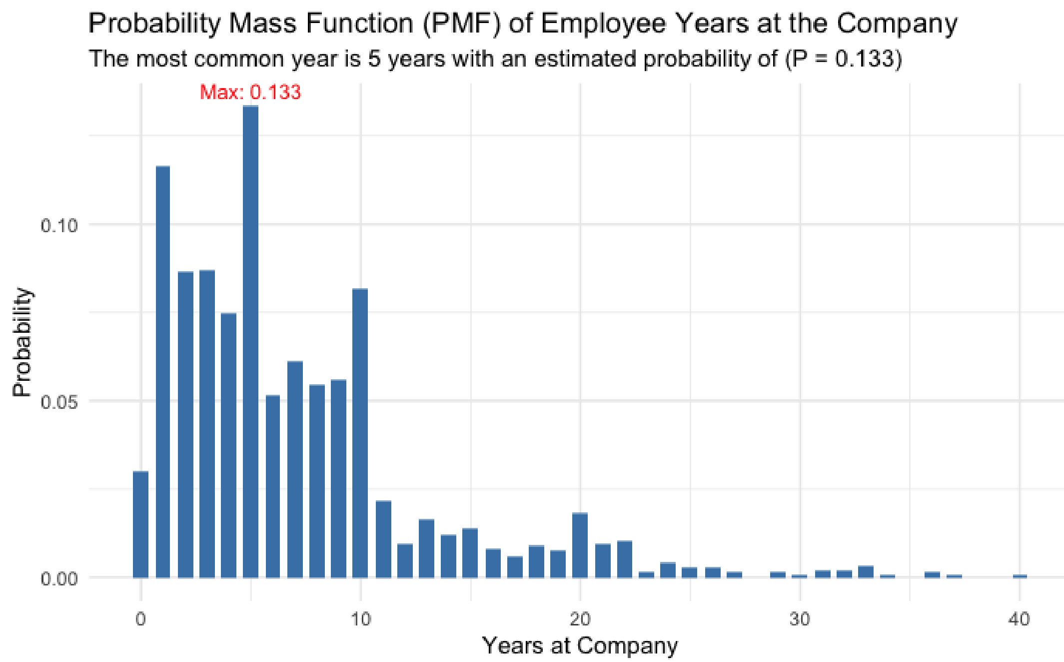
- $p(x)$gives the probability that the random variable X takes the value x.
- The PMF must satisfy:
  - $p(x) \geq 0$ for all x
  - $\sum_x p(x) = 1$

Table 8: PMF Table (First Few)

Interpretation: The table demonstrates that short-to-medium tenure **(0–5 years)** dominates the workforce, confirming that turnover or growth is concentrated in the first few years. The **peak at 5 years** may reflect internal HR structures, such as promotion cycles, contract renewals, or retention bonuses. The declining probability after 10 years highlights that long-term retention is relatively rare, suggesting the company may need strategies to support and retain experienced staff.

YearsAtCompany	Probability
0	3.0%
1	11.6%
2	8.6%
5	13.3%
10	8.2%

Figure 3: PMF of Years (as an Employee) at Company



Explanation: The plot shows the Probability Mass Function (PMF) of *Years at Company* for employees in the dataset. Each bar represents the probability that an employee has exactly a certain number of years of tenure. The total probability across all bars equals 1, as expected for a PMF.

From this plot, we could find that the distribution is right-skewed. Most employees cluster in the lower tenure values (0–10 years), with probabilities declining steadily for longer tenures. Only a small proportion of employees remain at the company beyond 20 years.

Interpretation: The highest probability occurs at 5 years, with an estimated probability of 0.133 (13.3%). This means that employees are most likely to have stayed for 5 years compared to any other specific duration. Moreover, nearly half of employees have **≤ 5 years of tenure** (as confirmed by the cumulative distribution function, CDF). This indicates that the workforce has a relatively short average tenure, possibly reflecting higher turnover or recent hiring growth. While the probabilities beyond 15 years drop to very low levels, showing that while some employees do remain for long periods (20+ years), they form only a small fraction of the total.

2) Methods: Mean (Expectation), Variance and Standard Deviation

Let X be a discrete random variable with possible values $x_1, x_2, \ldots, x_n$, and probability mass function (PMF) $P(X = x_i) = p_i$.

a) Mean (Expectation):

$$E[X] = \mu = \sum_{i=1}^{n} x_i \cdot p_i$$

b) Variance:

$$Var(X) = \sum_{i=1}^{n} (x_i - \mu)^2 \cdot p_i$$

Or equivalently,

$$Var(X) = E[X^2] - (E[X])^2,$$

where $E[X^2] = \sum_{i=1}^{n} x_i^2 \cdot p_i$

c) Standard Deviation:

$$SD(X) = \sqrt{Var(X)}$$

From the data and code output, we could find: The probability mass function reveals interesting tenure patterns. The highest probabilities occur at 1 year (11.6%) and 5 years (13.3%), suggesting two critical retention points. Many employees leave after their first year - likely those who find the role doesn't match expectations. The 5-year spike might represent employees who've gained enough experience to be attractive to competitors or who are reassessing their career trajectory. These aren't necessarily turnover points, but rather common tenure lengths in the current workforce.

Table 9: Mean, Variance and Standard Deviation Table of YearsAtCompany

Statistic	Value
Mean	7.01
Variance	37.53
Standard Deviation	6.13

Interpretation: The mean tenure of 7.01 years indicates a reasonably stable workforce, but the high variance (37.53) and standard deviation (6.13 years) reveal wide disparities in how long employees stay. This high variation means IBM has distinct employee segments - some leave quickly while others build long careers. The standard deviation being nearly as large as the mean suggests tenure doesn't follow a tight pattern but rather reflects diverse career paths within the company.

3) Cumulative Distribution Function (CDF):

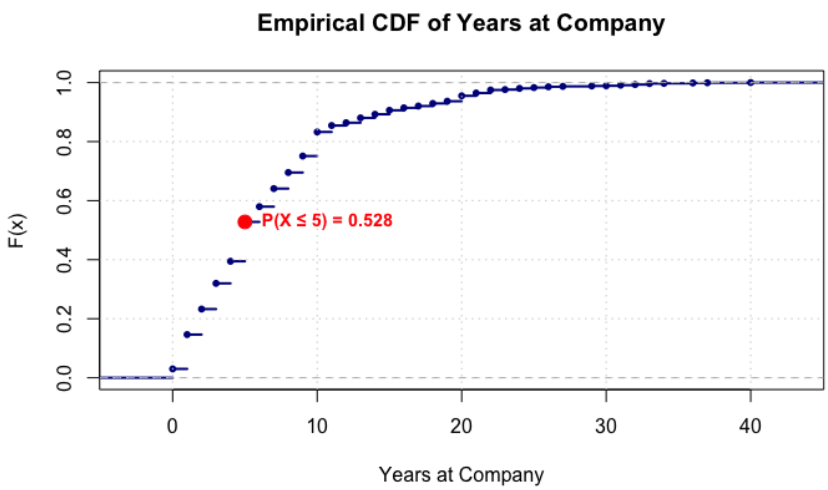
For a discrete random variable X, the cumulative distribution function is:

$$F(x) = P(X \leq x) = \sum_{t \leq x} p(t)$$

where:

- $F(x)$ gives the probability that X is less than or equal to x.
- The sum is over all possible values t of X such that $t \leq x$.

Figure 4: CDF of Years at Company



Interpretation: The CDF shows that employee tenure accumulates rapidly in early years then levels off. The steep rise through year 10 indicates most employees who leave do so within their first decade. After 10 years, the curve flattens significantly, suggesting employees who reach this milestone tend to stay much longer. This creates two distinct populations - shorter-term employees who cycle through relatively quickly and career employees who settle in for the long haul. The 52.8% mark at 5 years means over half the workforce has been with IBM for 5 years or less, indicating substantial regular turnover despite having some very long-tenured employees.

Therefore, the proportion of employees have been at the company ≤ 5 years is 52.8%, which means over half of the turnover employees have worked at the company for less than 5 years.

2. Continuous Random Variable: MonthlyIncome

1) Probability Density Function (PDF)

The probability density function (PDF) of a continuous random variable X, denoted as $f(x)$ describes the likelihood of X taking on a particular value. While f(x) itself is not a probability, the probability that X falls within an interval $[a, b]$ is given by:

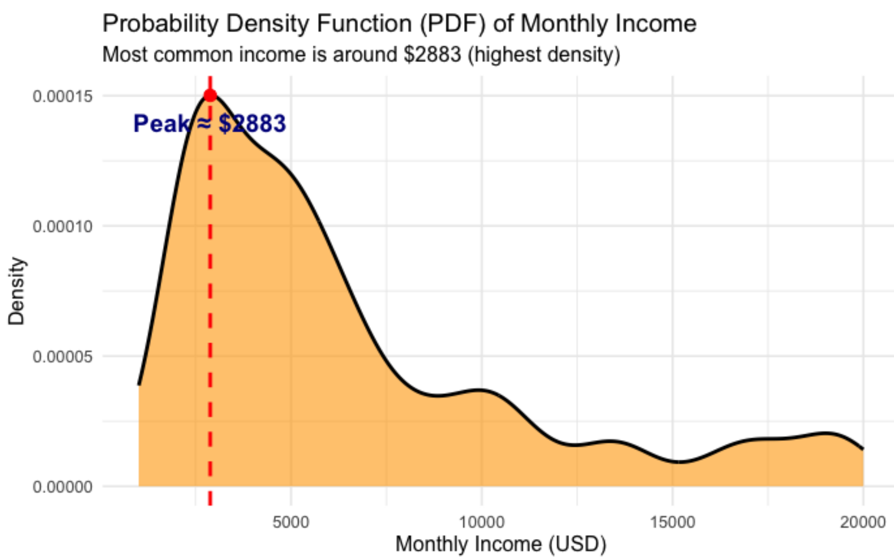
$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

The PDF must satisfy:

- $f(x) \geq 0$ for all x
- $\int_{-\infty}^{\infty} f(x)dx = 1$

Figure 5: Probability Density Function Plot of MonthlyIncome

We estimated the probability density function (PDF) of Monthly Income using kernel density estimation. As shown in the figure, for example, the estimated density at a monthly income of 4000 is approximately 0.00013. Note that this is a density, not a direct probability — probabilities over intervals are obtained by integrating the PDF.



Interpretation: The probability density function for monthly income shows a pronounced peak around $2883, indicating this is the most common salary range at IBM. The distribution has a long right tail extending to $20,000, confirming significant income inequality within the organization. The peak's sharpness suggests many employees are clustered in similar pay grades, likely representing standard compensation for common roles, while the tail represents specialized positions, senior management, and technical experts commanding premium salaries.

2) Methods: Mean (Expectation), Variance and Standard Deviation

a) Mean (Expectation):

The mean of a continuous random variable X with probability density function $f(x)$:

$$\mu = E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

b) Variance:

The variance of a continuous random variable X:

$$Var(X) = E[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)dx$$

Alternatively (using $E[X^2]$):

$$Var(X) = E[X^2] - (E[X])^2 = \int_{-\infty}^{\infty} x^2 f(x)dx - \left(\int_{-\infty}^{\infty} xf(x)dx\right)^2$$

c) Standard Deviation:

The standard deviation is the square root of variance:

$$\sigma = \sqrt{Var(X)}$$

Table 10: Mean, Variance and Standard Deviation Table of MonthlyIncome

Statistic	Value
Mean	6,503
Variance	22,164,857
Standard Deviation	4,708

Interpretation: The mean income of $6,503 masks considerable variation, with a variance of 22,164,857 and standard deviation of $4,708. This standard deviation being 72% of the mean indicates enormous pay disparities. Some employees earn barely above $1,000 while others approach $20,000 monthly. This variation likely reflects IBM's diverse workforce - from entry-level support staff to senior executives and specialized technical experts. Such wide compensation ranges can create challenges for employee satisfaction if not properly managed and explained.

3) Cumulative Distribution Function (CDF):

The CDF, $F(x)$ gives the probability that X is less than or equal to x:

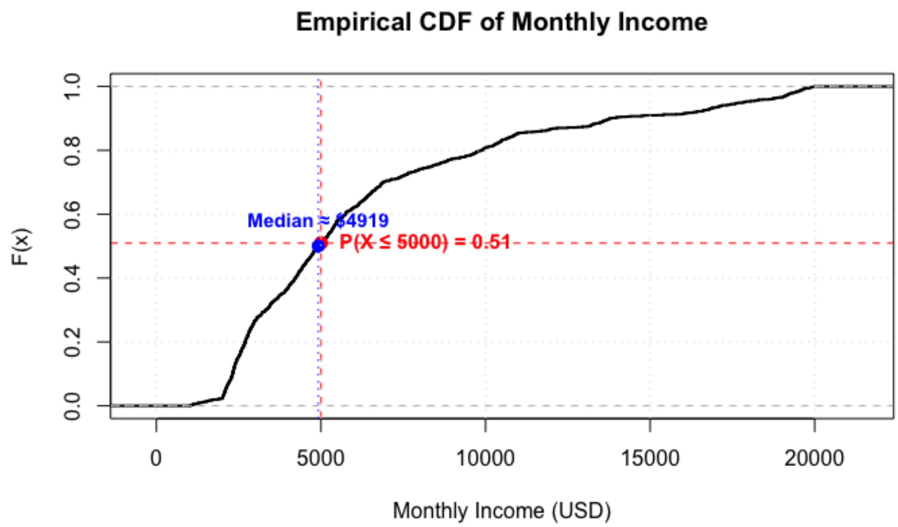
$$F(x) = P(X \leq x) = \int_{-\infty}^{x} xf(t)dt$$

where:

- $f(x)$ Probability density function (PDF) of X
- μ: Mean of X
- $Var(X)$: Variance

- σ: Standard deviation
- $F(x)$: Cumulative distribution function

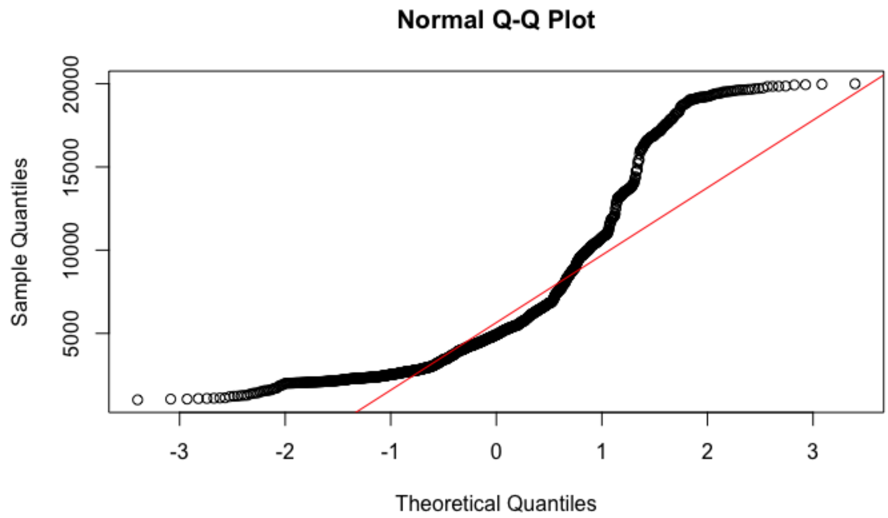
Figure 6: CDF of MonthlyIncome



Interpretation: The income CDF rises steeply through \$5,000 then gradually levels off. With 50.95% of employees earning \$5,000 or less, we see that the median employee earns modestly despite IBM's reputation as a major tech company. The gradual rise after \$5,000 indicates the higher-paid half of employees are spread across a wide range rather than clustering at any particular level. This shape - steep rise then long gradual tail - is typical for corporate compensation structures where many employees occupy standard roles while fewer reach senior positions.

Therefore, the proportion of employees earning ≤ \$5000 is 50.95%.

Figure 6: QQ Plot - Test for Normality



The QQ Plot above shows deviation from normality at both tails.

Interpretation: The QQ plot reveals that monthly income deviates significantly from a normal distribution. The points follow the reference line reasonably well in the middle range but curve away at both extremes. The upward curve at the right indicates a heavy right tail - there are more high earners than a normal distribution would predict. The flattening at the left shows income has a firm lower bound. This right-skewed pattern is expected for salary data where high earners stretch the distribution upward while minimum wage laws and cost-of-living create a floor below which salaries cannot fall.

Conclusion:

The analysis provides a comprehensive view of employee characteristics and workforce dynamics. Descriptive statistics revealed patterns in attrition, departmental composition, overtime, age, and income. Probability analysis showed that attrition is strongly linked to overtime and department, with higher risk in sales and among employees who work overtime. Random variable analysis demonstrated that tenure peaks at 5 years, while income is right-skewed, with most employees earning between $4,000 and $6,000. Together, these results illustrate how statistical methods can uncover both summary trends and deeper relationships in HR data.

Appendix

Carey Business School. *Statistical Analysis Module 1: Probability*. 2025.

Carey Business School. *Statistical Analysis Module 2: Discrete Random Variables*. 2024.

Sayah, Fares. *IBM HR Analytics: Employee Attrition & Performance*. Kaggle, 2022, https://www.kaggle.com/code/faressayah/ibm-hr-analytics-employee-attrition-performance.

Statistical_Analysis_Project

2025-09-06

Data Load

```r
# Load required packages
library(readr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(ggplot2)

# Read the CSV file
df <- read_csv("Statistical Analysis/Project/IBM_HR_Employee.csv")
```

```
## Rows: 1470 Columns: 35

## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (9): Attrition, BusinessTravel, Department, EducationField, Gender, Job...
## dbl (26): Age, DailyRate, DistanceFromHome, Education, EmployeeCount, Employ...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# View the structure of the data
str(df)
```

```
## spc_tbl_ [1,470 x 35] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ Age                     : num [1:1470] 41 49 37 33 27 32 59 30 38 36 ...
##  $ Attrition               : chr [1:1470] "Yes" "No" "Yes" "No" ...
##  $ BusinessTravel          : chr [1:1470] "Travel_Rarely" "Travel_Frequently" "Travel_Rarely" "Travel
##  $ DailyRate               : num [1:1470] 1102 279 1373 1392 591 ...
##  $ Department              : chr [1:1470] "Sales" "Research & Development" "Research & Development"
##  $ DistanceFromHome        : num [1:1470] 1 8 2 3 2 2 3 24 23 27 ...
##  $ Education               : num [1:1470] 2 1 2 4 1 2 3 1 3 3 ...
##  $ EducationField          : chr [1:1470] "Life Sciences" "Life Sciences" "Other" "Life Sciences" ..
##  $ EmployeeCount           : num [1:1470] 1 1 1 1 1 1 1 1 1 1 ...
##  $ EmployeeNumber          : num [1:1470] 1 2 4 5 7 8 10 11 12 13 ...
##  $ EnvironmentSatisfaction : num [1:1470] 2 3 4 4 1 4 3 4 4 3 ...
```

```
##  $ Gender                  : chr [1:1470] "Female" "Male" "Male" "Female" ...
##  $ HourlyRate              : num [1:1470] 94 61 92 56 40 79 81 67 44 94 ...
##  $ JobInvolvement          : num [1:1470] 3 2 2 3 3 3 4 3 2 3 ...
##  $ JobLevel                : num [1:1470] 2 2 1 1 1 1 1 1 3 2 ...
##  $ JobRole                 : chr [1:1470] "Sales Executive" "Research Scientist" "Laboratory Technici
##  $ JobSatisfaction         : num [1:1470] 4 2 3 3 2 4 1 3 3 3 ...
##  $ MaritalStatus           : chr [1:1470] "Single" "Married" "Single" "Married" ...
##  $ MonthlyIncome           : num [1:1470] 5993 5130 2090 2909 3468 ...
##  $ MonthlyRate             : num [1:1470] 19479 24907 2396 23159 16632 ...
##  $ NumCompaniesWorked      : num [1:1470] 8 1 6 1 9 0 4 1 0 6 ...
##  $ Over18                  : chr [1:1470] "Y" "Y" "Y" "Y" ...
##  $ OverTime                : chr [1:1470] "Yes" "No" "Yes" "Yes" ...
##  $ PercentSalaryHike       : num [1:1470] 11 23 15 11 12 13 20 22 21 13 ...
##  $ PerformanceRating       : num [1:1470] 3 4 3 3 3 3 4 4 4 3 ...
##  $ RelationshipSatisfaction: num [1:1470] 1 4 2 3 4 3 1 2 2 2 ...
##  $ StandardHours           : num [1:1470] 80 80 80 80 80 80 80 80 80 80 ...
##  $ StockOptionLevel        : num [1:1470] 0 1 0 0 1 0 3 1 0 2 ...
##  $ TotalWorkingYears       : num [1:1470] 8 10 7 8 6 8 12 1 10 17 ...
##  $ TrainingTimesLastYear   : num [1:1470] 0 3 3 3 3 2 3 2 2 3 ...
##  $ WorkLifeBalance         : num [1:1470] 1 3 3 3 3 2 2 3 3 2 ...
##  $ YearsAtCompany          : num [1:1470] 6 10 0 8 2 7 1 1 9 7 ...
##  $ YearsInCurrentRole      : num [1:1470] 4 7 0 7 2 7 0 0 7 7 ...
##  $ YearsSinceLastPromotion : num [1:1470] 0 1 0 3 2 3 0 0 1 7 ...
##  $ YearsWithCurrManager    : num [1:1470] 5 7 0 0 2 6 0 0 8 7 ...
##  - attr(*, "spec")=
##   .. cols(
##   ..   Age = col_double(),
##   ..   Attrition = col_character(),
##   ..   BusinessTravel = col_character(),
##   ..   DailyRate = col_double(),
##   ..   Department = col_character(),
##   ..   DistanceFromHome = col_double(),
##   ..   Education = col_double(),
##   ..   EducationField = col_character(),
##   ..   EmployeeCount = col_double(),
##   ..   EmployeeNumber = col_double(),
##   ..   EnvironmentSatisfaction = col_double(),
##   ..   Gender = col_character(),
##   ..   HourlyRate = col_double(),
##   ..   JobInvolvement = col_double(),
##   ..   JobLevel = col_double(),
##   ..   JobRole = col_character(),
##   ..   JobSatisfaction = col_double(),
##   ..   MaritalStatus = col_character(),
##   ..   MonthlyIncome = col_double(),
##   ..   MonthlyRate = col_double(),
##   ..   NumCompaniesWorked = col_double(),
##   ..   Over18 = col_character(),
##   ..   OverTime = col_character(),
##   ..   PercentSalaryHike = col_double(),
##   ..   PerformanceRating = col_double(),
##   ..   RelationshipSatisfaction = col_double(),
##   ..   StandardHours = col_double(),
##   ..   StockOptionLevel = col_double(),
```

```
## .. TotalWorkingYears = col_double(),
## .. TrainingTimesLastYear = col_double(),
## .. WorkLifeBalance = col_double(),
## .. YearsAtCompany = col_double(),
## .. YearsInCurrentRole = col_double(),
## .. YearsSinceLastPromotion = col_double(),
## .. YearsWithCurrManager = col_double()
## .. )
## - attr(*, "problems")=<externalptr>
```

```
# See first few rows
head(df)
```

```
## # A tibble: 6 x 35
##     Age Attrition BusinessTravel DailyRate Department DistanceFromHome Education
##   <dbl> <chr>     <chr>              <dbl> <chr>                 <dbl>     <dbl>
## 1    41 Yes       Travel_Rarely       1102 Sales                     1         2
## 2    49 No        Travel_Freque~       279 Research ~                8         1
## 3    37 Yes       Travel_Rarely       1373 Research ~                2         2
## 4    33 No        Travel_Freque~      1392 Research ~                3         4
## 5    27 No        Travel_Rarely        591 Research ~                2         1
## 6    32 No        Travel_Freque~      1005 Research ~                2         2
## # i 28 more variables: EducationField <chr>, EmployeeCount <dbl>,
## #   EmployeeNumber <dbl>, EnvironmentSatisfaction <dbl>, Gender <chr>,
## #   HourlyRate <dbl>, JobInvolvement <dbl>, JobLevel <dbl>, JobRole <chr>,
## #   JobSatisfaction <dbl>, MaritalStatus <chr>, MonthlyIncome <dbl>,
## #   MonthlyRate <dbl>, NumCompaniesWorked <dbl>, Over18 <chr>, OverTime <chr>,
## #   PercentSalaryHike <dbl>, PerformanceRating <dbl>,
## #   RelationshipSatisfaction <dbl>, StandardHours <dbl>, ...
```

Data Cleaning

```
# Remove rows with any missing values
df_clean <- df %>% na.omit()

# Check for duplicates
df_clean <- df_clean %>% distinct()

# Check unique values for key variables
unique(df_clean$Attrition)
```

```
## [1] "Yes" "No"
```

```
unique(df_clean$OverTime)
```

```
## [1] "Yes" "No"
```

```
unique(df_clean$Department)
```

```
## [1] "Sales"                  "Research & Development" "Human Resources"
```

Save the cleaned dataset as a new CSV file

```
write.csv(df_clean, "Statistical Analysis/Project/IBM_HR_Employee_Cleaned.csv", row.names = FALSE)
```

EDA

```r
# View dataset dimensions
dim(df_clean)
```

```
## [1] 1470   35
```

```r
# Quick summary of all variables
summary(df_clean)
```

```
##       Age         Attrition         BusinessTravel       DailyRate     
##  Min.   :18.00   Length:1470        Length:1470        Min.   : 102.0  
##  1st Qu.:30.00   Class :character   Class :character   1st Qu.: 465.0  
##  Median :36.00   Mode  :character   Mode  :character   Median : 802.0  
##  Mean   :36.92                                         Mean   : 802.5  
##  3rd Qu.:43.00                                         3rd Qu.:1157.0  
##  Max.   :60.00                                         Max.   :1499.0  
##   Department        DistanceFromHome   Education     EducationField    
##  Length:1470        Min.   : 1.000   Min.   :1.000   Length:1470       
##  Class :character   1st Qu.: 2.000   1st Qu.:2.000   Class :character  
##  Mode  :character   Median : 7.000   Median :3.000   Mode  :character  
##                     Mean   : 9.193   Mean   :2.913                     
##                     3rd Qu.:14.000   3rd Qu.:4.000                     
##                     Max.   :29.000   Max.   :5.000                     
##  EmployeeCount EmployeeNumber   EnvironmentSatisfaction    Gender         
##  Min.   :1     Min.   :   1.0   Min.   :1.000           Length:1470       
##  1st Qu.:1     1st Qu.: 491.2   1st Qu.:2.000           Class :character  
##  Median :1     Median :1020.5   Median :3.000           Mode  :character  
##  Mean   :1     Mean   :1024.9   Mean   :2.722                             
##  3rd Qu.:1     3rd Qu.:1555.8   3rd Qu.:4.000                             
##  Max.   :1     Max.   :2068.0   Max.   :4.000                             
##    HourlyRate     JobInvolvement    JobLevel       JobRole         
##  Min.   : 30.00   Min.   :1.00   Min.   :1.000   Length:1470       
##  1st Qu.: 48.00   1st Qu.:2.00   1st Qu.:1.000   Class :character  
##  Median : 66.00   Median :3.00   Median :2.000   Mode  :character  
##  Mean   : 65.89   Mean   :2.73   Mean   :2.064                     
##  3rd Qu.: 83.75   3rd Qu.:3.00   3rd Qu.:3.000                     
##  Max.   :100.00   Max.   :4.00   Max.   :5.000                     
##  JobSatisfaction MaritalStatus      MonthlyIncome    MonthlyRate   
##  Min.   :1.000   Length:1470        Min.   : 1009   Min.   : 2094  
##  1st Qu.:2.000   Class :character   1st Qu.: 2911   1st Qu.: 8047  
##  Median :3.000   Mode  :character   Median : 4919   Median :14236  
##  Mean   :2.729                      Mean   : 6503   Mean   :14313  
##  3rd Qu.:4.000                      3rd Qu.: 8379   3rd Qu.:20462  
##  Max.   :4.000                      Max.   :19999   Max.   :26999  
##  NumCompaniesWorked    Over18             OverTime         PercentSalaryHike
##  Min.   :0.000      Length:1470        Length:1470        Min.   :11.00    
##  1st Qu.:1.000      Class :character   Class :character   1st Qu.:12.00    
##  Median :2.000      Mode  :character   Mode  :character   Median :14.00    
##  Mean   :2.693                                            Mean   :15.21    
##  3rd Qu.:4.000                                            3rd Qu.:18.00    
##  Max.   :9.000                                            Max.   :25.00    
##  PerformanceRating RelationshipSatisfaction StandardHours StockOptionLevel
##  Min.   :3.000     Min.   :1.000            Min.   :80    Min.   :0.0000  
##  1st Qu.:3.000     1st Qu.:2.000            1st Qu.:80    1st Qu.:0.0000  
```

```
##  Median :3.000     Median :3.000            Median :80    Median :1.0000
##  Mean   :3.154     Mean   :2.712            Mean   :80    Mean   :0.7939
##  3rd Qu.:3.000     3rd Qu.:4.000            3rd Qu.:80    3rd Qu.:1.0000
##  Max.   :4.000     Max.   :4.000            Max.   :80    Max.   :3.0000
##  TotalWorkingYears TrainingTimesLastYear WorkLifeBalance YearsAtCompany
##  Min.   : 0.00     Min.   :0.000         Min.   :1.000   Min.   : 0.000
##  1st Qu.: 6.00     1st Qu.:2.000         1st Qu.:2.000   1st Qu.: 3.000
##  Median :10.00     Median :3.000         Median :3.000   Median : 5.000
##  Mean   :11.28     Mean   :2.799         Mean   :2.761   Mean   : 7.008
##  3rd Qu.:15.00     3rd Qu.:3.000         3rd Qu.:3.000   3rd Qu.: 9.000
##  Max.   :40.00     Max.   :6.000         Max.   :4.000   Max.   :40.000
##  YearsInCurrentRole YearsSinceLastPromotion YearsWithCurrManager
##  Min.   : 0.000     Min.   : 0.000          Min.   : 0.000
##  1st Qu.: 2.000     1st Qu.: 0.000          1st Qu.: 2.000
##  Median : 3.000     Median : 1.000          Median : 3.000
##  Mean   : 4.229     Mean   : 2.188          Mean   : 4.123
##  3rd Qu.: 7.000     3rd Qu.: 3.000          3rd Qu.: 7.000
##  Max.   :18.000     Max.   :15.000          Max.   :17.000
```

```r
# See the first 10 rows
head(df_clean, 10)
```

```
## # A tibble: 10 x 35
##      Age Attrition BusinessTravel    DailyRate Department        DistanceFromHome
##    <dbl> <chr>     <chr>                 <dbl> <chr>                        <dbl>
##  1    41 Yes       Travel_Rarely          1102 Sales                            1
##  2    49 No        Travel_Frequently       279 Research & Deve~                 8
##  3    37 Yes       Travel_Rarely          1373 Research & Deve~                 2
##  4    33 No        Travel_Frequently      1392 Research & Deve~                 3
##  5    27 No        Travel_Rarely           591 Research & Deve~                 2
##  6    32 No        Travel_Frequently      1005 Research & Deve~                 2
##  7    59 No        Travel_Rarely          1324 Research & Deve~                 3
##  8    30 No        Travel_Rarely          1358 Research & Deve~                24
##  9    38 No        Travel_Frequently       216 Research & Deve~                23
## 10    36 No        Travel_Rarely          1299 Research & Deve~                27
## # i 29 more variables: Education <dbl>, EducationField <chr>,
## #   EmployeeCount <dbl>, EmployeeNumber <dbl>, EnvironmentSatisfaction <dbl>,
## #   Gender <chr>, HourlyRate <dbl>, JobInvolvement <dbl>, JobLevel <dbl>,
## #   JobRole <chr>, JobSatisfaction <dbl>, MaritalStatus <chr>,
## #   MonthlyIncome <dbl>, MonthlyRate <dbl>, NumCompaniesWorked <dbl>,
## #   Over18 <chr>, OverTime <chr>, PercentSalaryHike <dbl>,
## #   PerformanceRating <dbl>, RelationshipSatisfaction <dbl>, ...
```

```r
# Check variable names
colnames(df_clean)
```

```
##  [1] "Age"                      "Attrition"
##  [3] "BusinessTravel"           "DailyRate"
##  [5] "Department"               "DistanceFromHome"
##  [7] "Education"                "EducationField"
##  [9] "EmployeeCount"            "EmployeeNumber"
## [11] "EnvironmentSatisfaction"  "Gender"
## [13] "HourlyRate"               "JobInvolvement"
## [15] "JobLevel"                 "JobRole"
## [17] "JobSatisfaction"          "MaritalStatus"
## [19] "MonthlyIncome"            "MonthlyRate"
```

```
## [21] "NumCompaniesWorked"       "Over18"
## [23] "OverTime"                 "PercentSalaryHike"
## [25] "PerformanceRating"        "RelationshipSatisfaction"
## [27] "StandardHours"            "StockOptionLevel"
## [29] "TotalWorkingYears"        "TrainingTimesLastYear"
## [31] "WorkLifeBalance"          "YearsAtCompany"
## [33] "YearsInCurrentRole"       "YearsSinceLastPromotion"
## [35] "YearsWithCurrManager"
```

Categorical Variables

```r
# Identify categorical variable names
cat_vars <- c("Attrition", "OverTime", "Department", "Gender", "EducationField", "JobRole",
              "MaritalStatus", "BusinessTravel")


df_clean[cat_vars] <- lapply(df_clean[cat_vars], as.factor)

# Check all the selected are categorical variables
# str(df_clean[cat_vars])
```

Check for Missing Values

```r
colSums(is.na(df_clean))
```

```
##                      Age                Attrition           BusinessTravel
##                        0                        0                        0
##                DailyRate               Department         DistanceFromHome
##                        0                        0                        0
##                Education           EducationField            EmployeeCount
##                        0                        0                        0
##           EmployeeNumber  EnvironmentSatisfaction                   Gender
##                        0                        0                        0
##               HourlyRate           JobInvolvement                 JobLevel
##                        0                        0                        0
##                  JobRole          JobSatisfaction            MaritalStatus
##                        0                        0                        0
##            MonthlyIncome              MonthlyRate       NumCompaniesWorked
##                        0                        0                        0
##                   Over18                 OverTime        PercentSalaryHike
##                        0                        0                        0
##        PerformanceRating RelationshipSatisfaction            StandardHours
##                        0                        0                        0
##         StockOptionLevel        TotalWorkingYears    TrainingTimesLastYear
##                        0                        0                        0
##          WorkLifeBalance           YearsAtCompany       YearsInCurrentRole
##                        0                        0                        0
##  YearsSinceLastPromotion     YearsWithCurrManager
##                        0                        0
```

Module 0 - Descriptive Statistics Report

The dataset used in this project is the IBM HR Employee Attrition dataset. It contains information on 1,470 employees, with 35 variables including demographic information, job roles, compensation, work environment, and attrition status. A full variable list is included in the appendix. In this report, we focus on a selected group of variables that are most relevant for both business analysis and the statistical concepts covered in this course.

```
# Select key variables for Module 0
df_selected <- df %>%
  select(Attrition, Department, OverTime, Age, MonthlyIncome, YearsAtCompany)

# View the structure
str(df_selected)
```

```
## tibble [1,470 x 6] (S3: tbl_df/tbl/data.frame)
##  $ Attrition     : chr [1:1470] "Yes" "No" "Yes" "No" ...
##  $ Department    : chr [1:1470] "Sales" "Research & Development" "Research & Development" "Research &
##  $ OverTime      : chr [1:1470] "Yes" "No" "Yes" "Yes" ...
##  $ Age           : num [1:1470] 41 49 37 33 27 32 59 30 38 36 ...
##  $ MonthlyIncome : num [1:1470] 5993 5130 2090 2909 3468 ...
##  $ YearsAtCompany: num [1:1470] 6 10 0 8 2 7 1 1 9 7 ...
```

```
summary(df_selected)
```

```
##   Attrition          Department          OverTime              Age
##  Length:1470        Length:1470        Length:1470        Min.   :18.00
##  Class :character   Class :character   Class :character   1st Qu.:30.00
##  Mode  :character   Mode  :character   Mode  :character   Median :36.00
##                                                           Mean   :36.92
##                                                           3rd Qu.:43.00
##                                                           Max.   :60.00
##  MonthlyIncome   YearsAtCompany
##  Min.   : 1009   Min.   : 0.000
##  1st Qu.: 2911   1st Qu.: 3.000
##  Median : 4919   Median : 5.000
##  Mean   : 6503   Mean   : 7.008
##  3rd Qu.: 8379   3rd Qu.: 9.000
##  Max.   :19999   Max.   :40.000
```

```
head(df_selected)
```

```
## # A tibble: 6 x 6
##   Attrition Department             OverTime   Age MonthlyIncome YearsAtCompany
##   <chr>     <chr>                  <chr>    <dbl>         <dbl>          <dbl>
## 1 Yes       Sales                  Yes         41          5993              6
## 2 No        Research & Development No          49          5130             10
## 3 Yes       Research & Development Yes         37          2090              0
## 4 No        Research & Development Yes         33          2909              8
## 5 No        Research & Development No          27          3468              2
## 6 No        Research & Development No          32          3068              7
```

Categorical Variables

```
# Attrition
table(df_clean$Attrition)
```

```
##
##    No  Yes
## 1233  237
```

```
prop.table(table(df_clean$Attrition))
```

```
##
##        No       Yes
## 0.8387755 0.1612245
```

```
# Department
table(df_clean$Department)
```

```
##
##        Human Resources Research & Development                  Sales
##                     63                    961                    446
```

```
prop.table(table(df_clean$Department))
```

```
##
##        Human Resources Research & Development                  Sales
##             0.04285714             0.65374150             0.30340136
```

```
# OverTime
table(df_clean$OverTime)
```

```
##
##    No  Yes
## 1054  416
```

```
prop.table(table(df_clean$OverTime))
```

```
##
##        No       Yes
## 0.7170068 0.2829932
```

Numerical Variables

```
# Age, MonthlyIncome, YearsAtCompany

summary(df_clean$Age)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   18.00   30.00   36.00   36.92   43.00   60.00
```

```
summary(df_clean$MonthlyIncome)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1009    2911    4919    6503    8379   19999
```

```
summary(df_clean$YearsAtCompany)
```

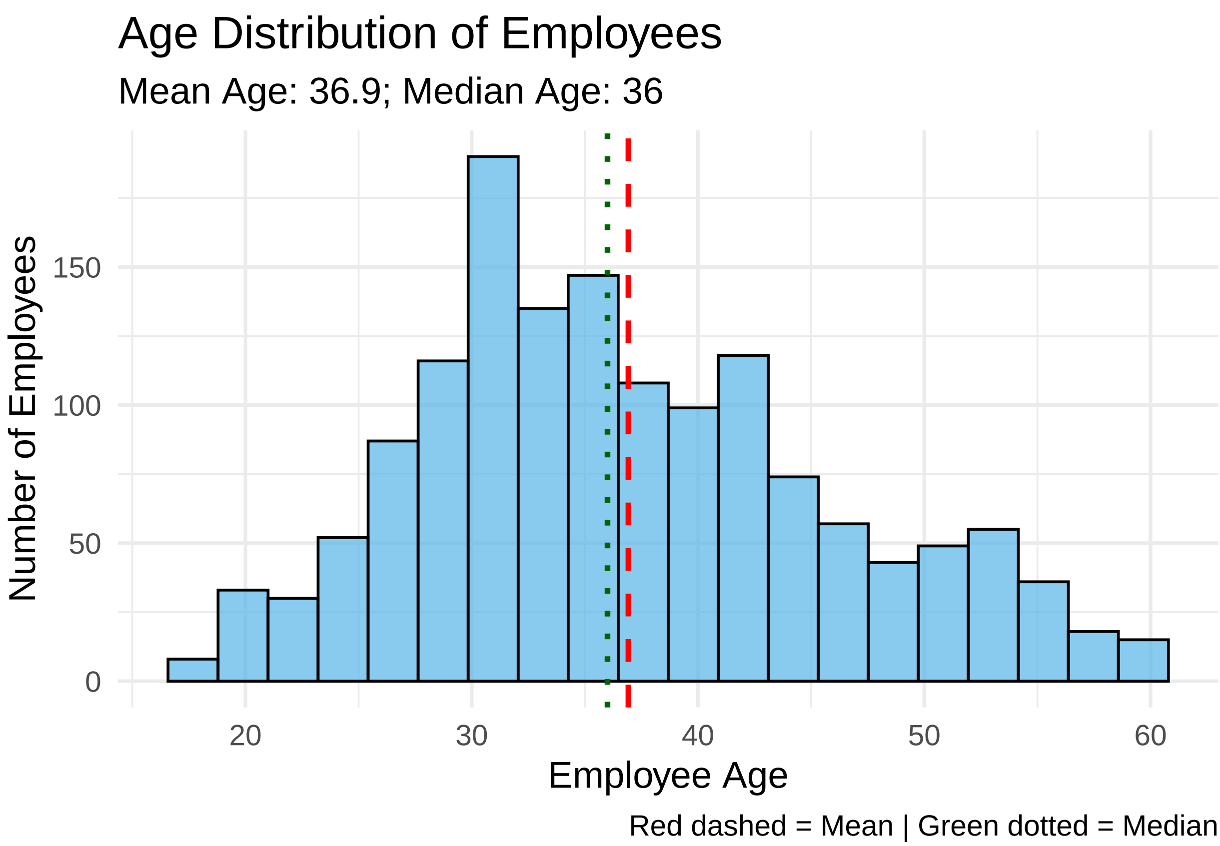
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.000   3.000   5.000   7.008   9.000  40.000
```

Visualization

```
library(ggplot2)
library(dplyr)
```

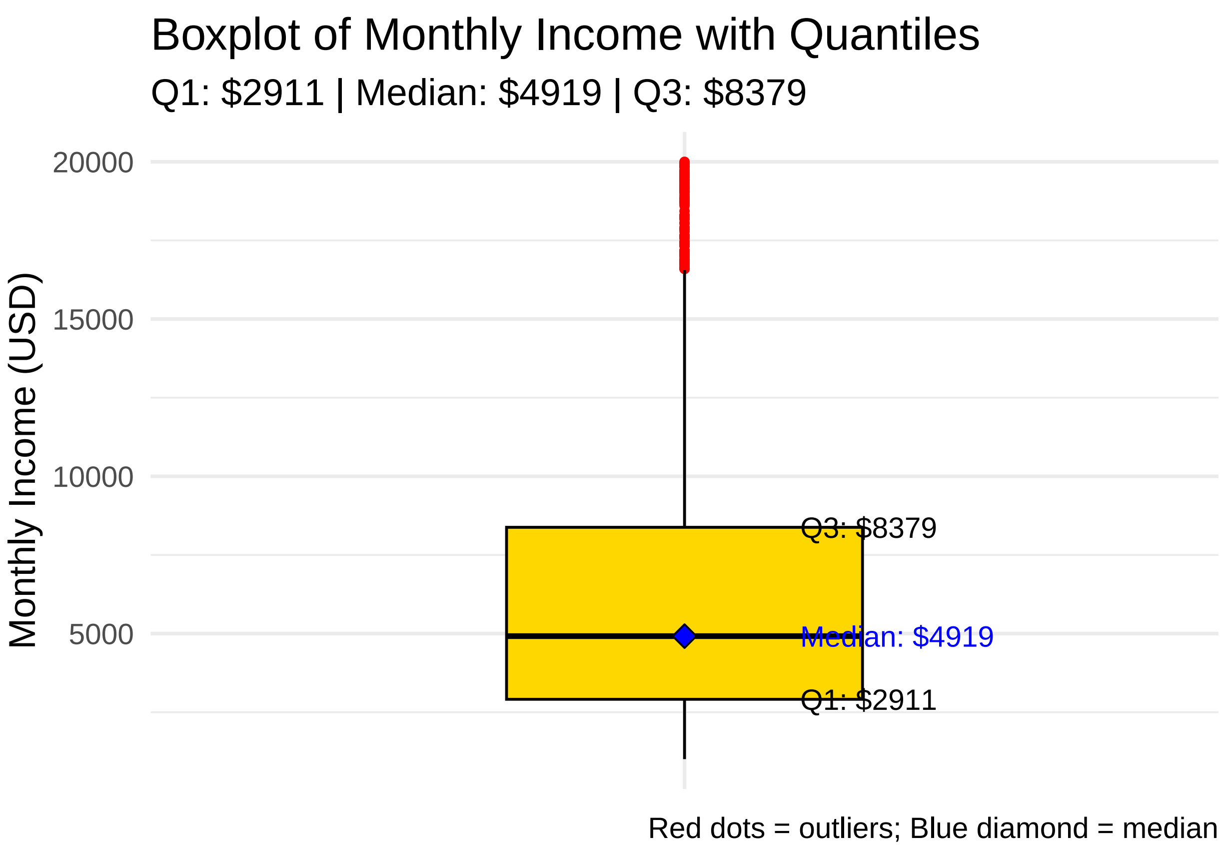
```r
# Calculate mean and median age
mean_age <- mean(df_clean$Age)
median_age <- median(df_clean$Age)

ggplot(df_clean, aes(x = Age)) +
  geom_histogram(bins = 20, fill = "#56B4E9", color = "black", alpha = 0.7) +
  geom_vline(aes(xintercept = mean_age), color = "red", linetype = "dashed", size = 1, show.legend = TRU
  geom_vline(aes(xintercept = median_age), color = "darkgreen", linetype = "dotted", size = 1, show.lege
  labs(
    title = "Age Distribution of Employees",
    subtitle = paste0("Mean Age: ", round(mean_age, 1), "; Median Age: ", median_age),
    x = "Employee Age",
    y = "Number of Employees",
    caption = "Red dashed = Mean | Green dotted = Median"
  ) +
  theme_minimal(base_size = 14)
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```r
# Compute quantiles
q1 <- quantile(df_clean$MonthlyIncome, 0.25)
med <- quantile(df_clean$MonthlyIncome, 0.5)
q3 <- quantile(df_clean$MonthlyIncome, 0.75)
```

```r
ggplot(df_clean, aes(x = "", y = MonthlyIncome)) +
  geom_boxplot(fill = "gold", color = "black", width = 0.4, outlier.color = "red", outlier.shape = 16) +
  stat_summary(fun = median, geom = "point", shape = 23, size = 3, fill = "blue", color = "black") +

  annotate("text", x = 1.13, y = q1, label = paste0("Q1: $", q1), hjust = 0, color = "black", size = 4)
  annotate("text", x = 1.13, y = med, label = paste0("Median: $", med), hjust = 0, color = "blue", size
  annotate("text", x = 1.13, y = q3, label = paste0("Q3: $", q3), hjust = 0, color = "black", size = 4)
  labs(
    title = "Boxplot of Monthly Income with Quantiles",
    subtitle = paste0("Q1: $", q1, " | Median: $", med, " | Q3: $", q3),
    y = "Monthly Income (USD)",
    x = NULL,
    caption = "Red dots = outliers; Blue diamond = median"
  ) +
  theme_minimal(base_size = 14) +
  theme(axis.text.x = element_blank(), axis.ticks.x = element_blank())
```




```r
# Relationships between Variables

table(df_clean$Attrition, df_clean$Department)
```

```
##
##       Human Resources Research & Development Sales
##   No               51                    828   354
##   Yes              12                    133    92
```

```r
prop.table(table(df_clean$Attrition, df_clean$Department), 2)
```

```
##
##       Human Resources Research & Development     Sales
##   No        0.8095238              0.8616025 0.7937220
##   Yes       0.1904762              0.1383975 0.2062780
```

Module 1 - Probability Analysis

Let's use the variables: Attrition (Yes/No), OverTime (Yes/No).

Attrition vs OverTime

Joint Probability

```r
# Create contingency table
joint_table <- table(df_selected$Attrition, df_selected$OverTime)
joint_table
```

```
##
##        No Yes
##   No  944 289
##   Yes 110 127
```

```r
# Joint probabilities
joint_prob <- prop.table(joint_table)
joint_prob
```

```
##
##                No        Yes
##   No  0.64217687 0.19659864
##   Yes 0.07482993 0.08639456
```

Conditional Probability

```r
# Conditional probability: P(Attrition | OverTime)
cond_prob_attrition_given_overtime <- prop.table(joint_table, 2)
cond_prob_attrition_given_overtime
```

```
##
##               No       Yes
##   No  0.8956357 0.6947115
##   Yes 0.1043643 0.3052885
```

```r
# Conditional probability: P(OverTime | Attrition)
cond_prob_overtime_given_attrition <- prop.table(joint_table, 1)
cond_prob_overtime_given_attrition
```

```
##
##               No       Yes
##   No  0.7656123 0.2343877
##   Yes 0.4641350 0.5358650
```

Bayes' Theorem

Let's compute: P(Attrition=Yes | OverTime=Yes) using Bayes' theorem.

Formula:

```r
# Calculate P(OverTime="Yes" | Attrition="Yes")
p_overtime_yes_given_attrition_yes <- joint_table["Yes", "Yes"] / sum(joint_table["Yes", ])

# Calculate P(Attrition="Yes")
p_attrition_yes <- sum(df_selected$Attrition == "Yes") / nrow(df_selected)

# Calculate P(OverTime="Yes")
p_overtime_yes <- sum(df_selected$OverTime == "Yes") / nrow(df_selected)

# Bayes' theorem
p_attrition_yes_given_overtime_yes <- (p_overtime_yes_given_attrition_yes * p_attrition_yes) / p_overti
p_attrition_yes_given_overtime_yes
```

```
## [1] 0.3052885
```

Attrition vs. Department

Try more variables

```r
# 1. Create contingency table
attr_dept_table <- table(df_selected$Attrition, df_selected$Department)
attr_dept_table
```

```
##
##       Human Resources Research & Development Sales
##   No               51                    828   354
##   Yes              12                    133    92
```

```r
# 2. Joint probability table
attr_dept_joint_prob <- prop.table(attr_dept_table)
attr_dept_joint_prob
```

```
##
##       Human Resources Research & Development       Sales
##   No      0.034693878            0.563265306 0.240816327
##   Yes     0.008163265            0.090476190 0.062585034
```

```r
# 3. Conditional probability: P(Attrition | Department)
# (Probability of attrition status within each department)
cond_prob_attr_given_dept <- prop.table(attr_dept_table, 2)
cond_prob_attr_given_dept
```

```
##
##       Human Resources Research & Development     Sales
##   No        0.8095238              0.8616025 0.7937220
##   Yes       0.1904762              0.1383975 0.2062780
```

```r
# 4. Conditional probability: P(Department | Attrition)
# (Probability of department within each attrition group)
cond_prob_dept_given_attr <- prop.table(attr_dept_table, 1)
cond_prob_dept_given_attr
```

```
##
##       Human Resources Research & Development      Sales
##   No       0.04136253             0.67153285 0.28710462
##   Yes      0.05063291             0.56118143 0.38818565
```

Bayes' Theorem

P(Attrition = Yes | Department = Sales) using Bayes' theorem:

```
# P(Department = Sales | Attrition = Yes)
p_dept_sales_given_attr_yes <- attr_dept_table["Yes", "Sales"] / sum(attr_dept_table["Yes", ])
# P(Attrition = Yes)
p_attr_yes <- sum(df_selected$Attrition == "Yes") / nrow(df_selected)
# P(Department = Sales)
p_dept_sales <- sum(df_selected$Department == "Sales") / nrow(df_selected)

# Apply Bayes' theorem
p_attr_yes_given_dept_sales <- (p_dept_sales_given_attr_yes * p_attr_yes) / p_dept_sales
p_attr_yes_given_dept_sales
```

```
## [1] 0.206278
```

Module 2 - Random Variables, PMF, Expectation, Variance, and CDF

Define the Random Variable - Discrete

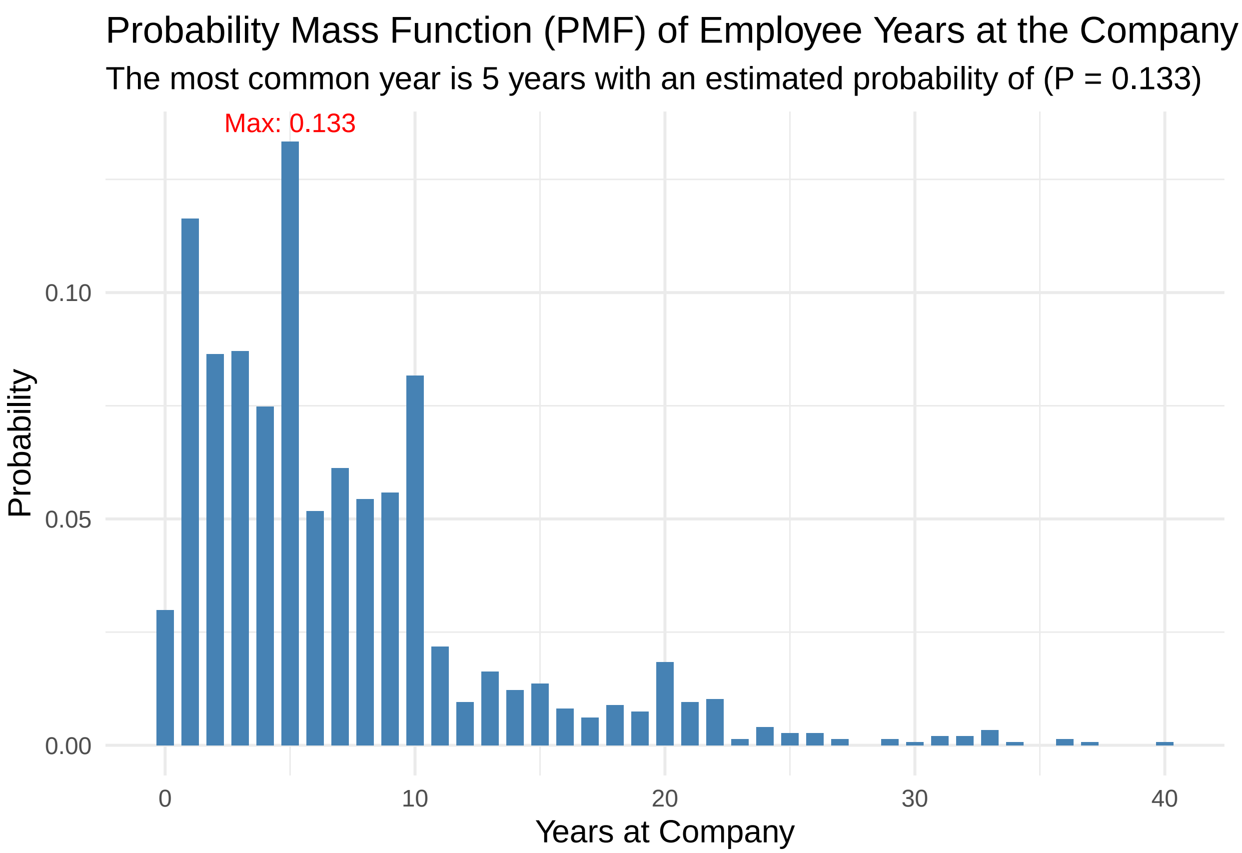
Let X = "YearsAtCompany" for a randomly selected employee, which is a discrete random variable (showing how many years an employee has worked at the company).

```
# Get the random variable
X <- df_selected$YearsAtCompany

# PMF: Frequency table of YearsAtCompany
pmf_table <- table(X) / length(X)
pmf_table
```

```
## X
##            0            1            2            3            4            5
## 0.0299319728 0.1163265306 0.0863945578 0.0870748299 0.0748299320 0.1333333333
##            6            7            8            9           10           11
## 0.0517006803 0.0612244898 0.0544217687 0.0557823129 0.0816326531 0.0217687075
##           12           13           14           15           16           17
## 0.0095238095 0.0163265306 0.0122448980 0.0136054422 0.0081632653 0.0061224490
##           18           19           20           21           22           23
## 0.0088435374 0.0074829932 0.0183673469 0.0095238095 0.0102040816 0.0013605442
##           24           25           26           27           29           30
## 0.0040816327 0.0027210884 0.0027210884 0.0013605442 0.0013605442 0.0006802721
##           31           32           33           34           36           37
## 0.0020408163 0.0020408163 0.0034013605 0.0006802721 0.0013605442 0.0006802721
##           40
## 0.0006802721
```

```
# Convert to a data frame for plotting
pmf_df <- as.data.frame(pmf_table)
colnames(pmf_df) <- c("YearsAtCompany", "Probability")


# Plot the PMF
library(ggplot2)
```

```r
# Find the max
max_point <- pmf_df[which.max(pmf_df$Probability), ]

ggplot(pmf_df, aes(x = as.numeric(as.character(YearsAtCompany)), y = Probability)) +
  geom_col(fill = "steelblue", width = 0.7) +

  geom_text(data = max_point,
            aes(label = paste0("Max: ", round(Probability, 3))),
            vjust = -0.5, size = 3.5, color = "red") +

  labs(
    title = "Probability Mass Function (PMF) of Employee Years at the Company",
    subtitle = paste0("The most common year is ", max_point$YearsAtCompany,
                      " years with an estimated probability of (P = ", round(max_point$Probability, 3),
    x = "Years at Company",
    y = "Probability"
  ) +
  theme_minimal(base_size = 11.5)
```



Expectation (Mean) and Variance

```r
# Expectation (mean)
mean_X <- mean(X)
mean_X
```

```
## [1] 7.008163
```

```r
# Variance
var_X <- var(X)
var_X
```

```
## [1] 37.53431
```

```r
# Standard Deviation
sd_X <- sd(X)
sd_X
```

```
## [1] 6.126525
```

Cumulative Distribution Function (CDF)

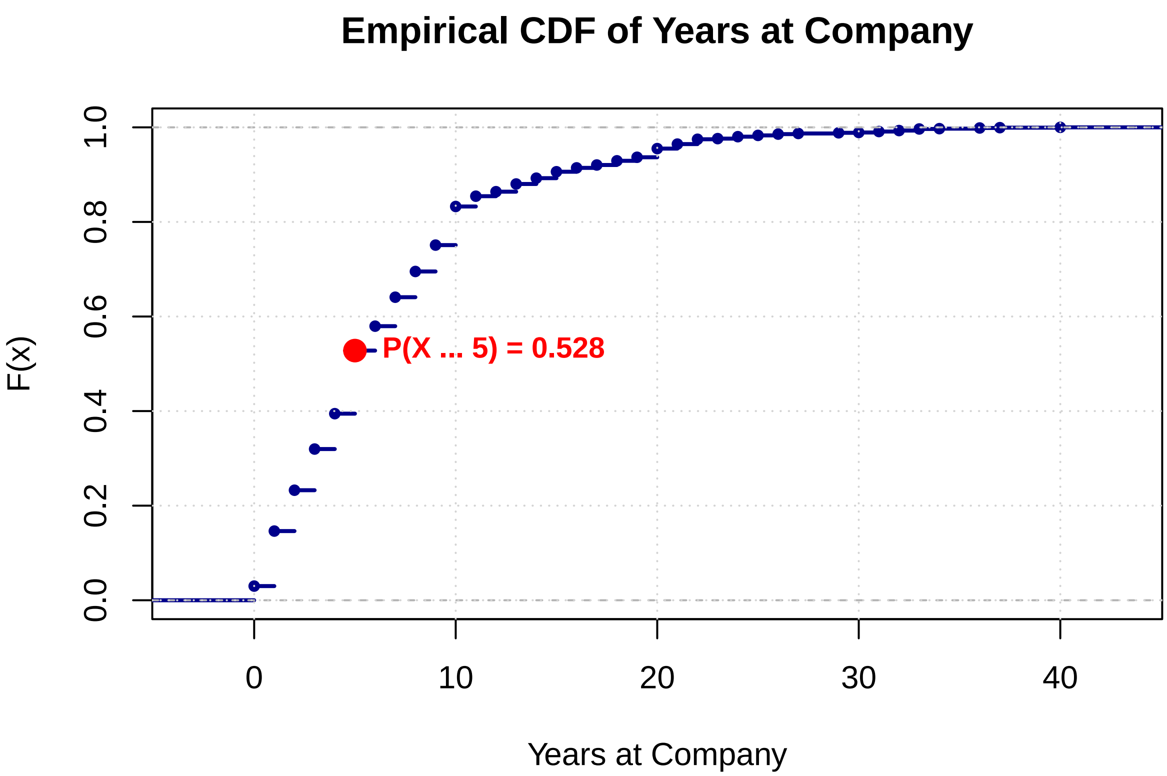
```r
# Empirical CDF
cdf_X <- ecdf(X)

# Plot the CDF
# plot(cdf_X, main="Empirical CDF of Years at Company", xlab="Years at Company", ylab="F(x)")

# What proportion of employees have been at the company   5 years?
cdf_X(5)
```

```
## [1] 0.5278912
```

```r
# Empirical CDF
cdf_X <- ecdf(X)

# Compute key probability: P(X   5)
p_le5 <- cdf_X(5)

# Plot the CDF
plot(cdf_X,
     main = "Empirical CDF of Years at Company",
     xlab = "Years at Company",
     ylab = "F(x)",
     col = "darkblue", lwd = 2, pch = 16, cex = 0.8)

# Highlight the point at X = 5
points(5, p_le5, col = "red", pch = 19, cex = 1.5)
text(5, p_le5, labels = paste0(" P(X   5) = ", round(p_le5, 3)),
     pos = 4, col = "red", cex = 0.9, font = 2)
```

```
## Warning in text.default(5, p_le5, labels = paste0(" P(X   5) = ", round(p_le5,
## : conversion failure on ' P(X   5) = 0.528' in 'mbcsToSbcs': dot substituted
## for <e2>

## Warning in text.default(5, p_le5, labels = paste0(" P(X   5) = ", round(p_le5,
## : conversion failure on ' P(X   5) = 0.528' in 'mbcsToSbcs': dot substituted
## for <89>

## Warning in text.default(5, p_le5, labels = paste0(" P(X   5) = ", round(p_le5,
## : conversion failure on ' P(X   5) = 0.528' in 'mbcsToSbcs': dot substituted
## for <a4>
```

```r
# Add grid for readability
grid()
```



Define the Random Variable - Continuous

Let Y = "MonthlyIncome" for a randomly selected employee, which is a continuous random variable, showing the monthly income of employees.

Probability Density Function (PDF)

```
density_income <- density(df_selected$MonthlyIncome)
head(density_income$x)
```

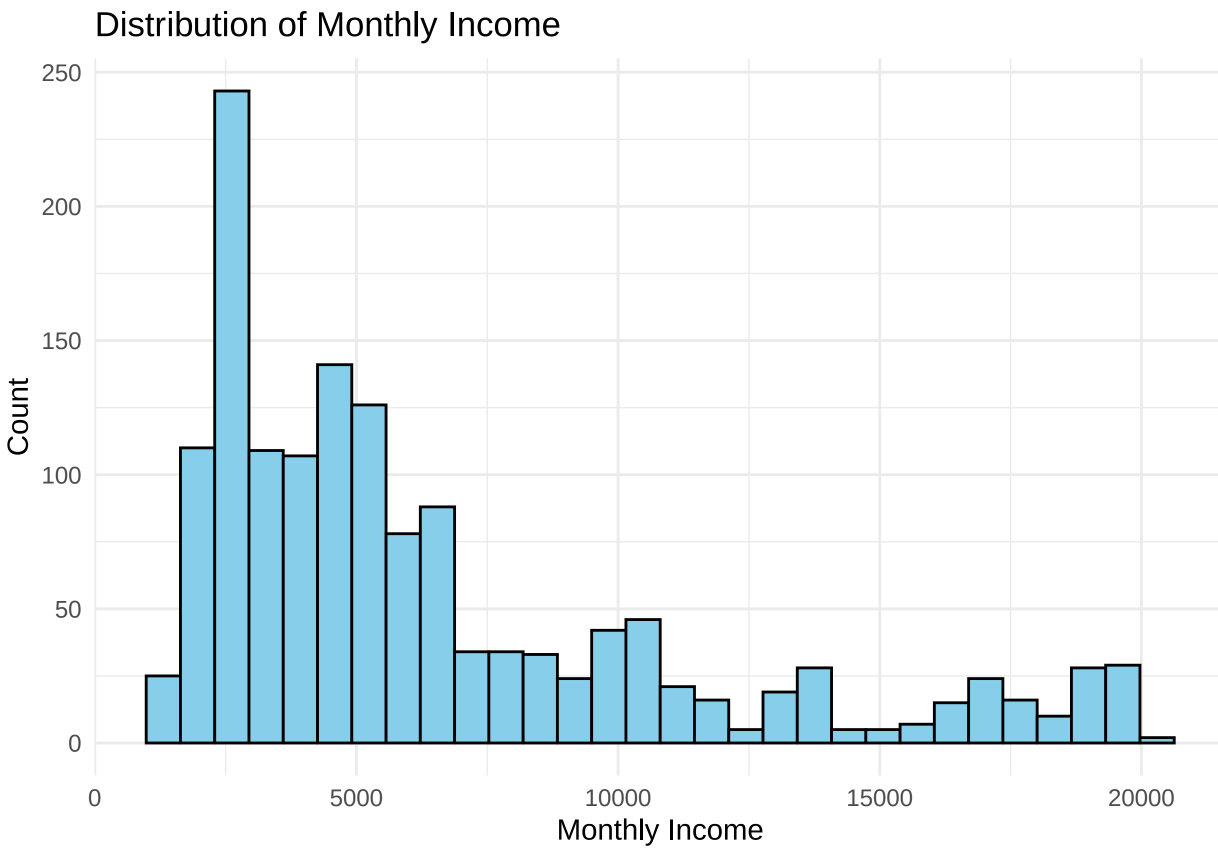
```
## [1] -1553.265 -1506.074 -1458.883 -1411.692 -1364.501 -1317.310
```

```
head(density_income$y)
```

```
## [1] 4.361069e-08 5.221756e-08 6.230126e-08 7.465970e-08 8.904824e-08
## [6] 1.057140e-07
```

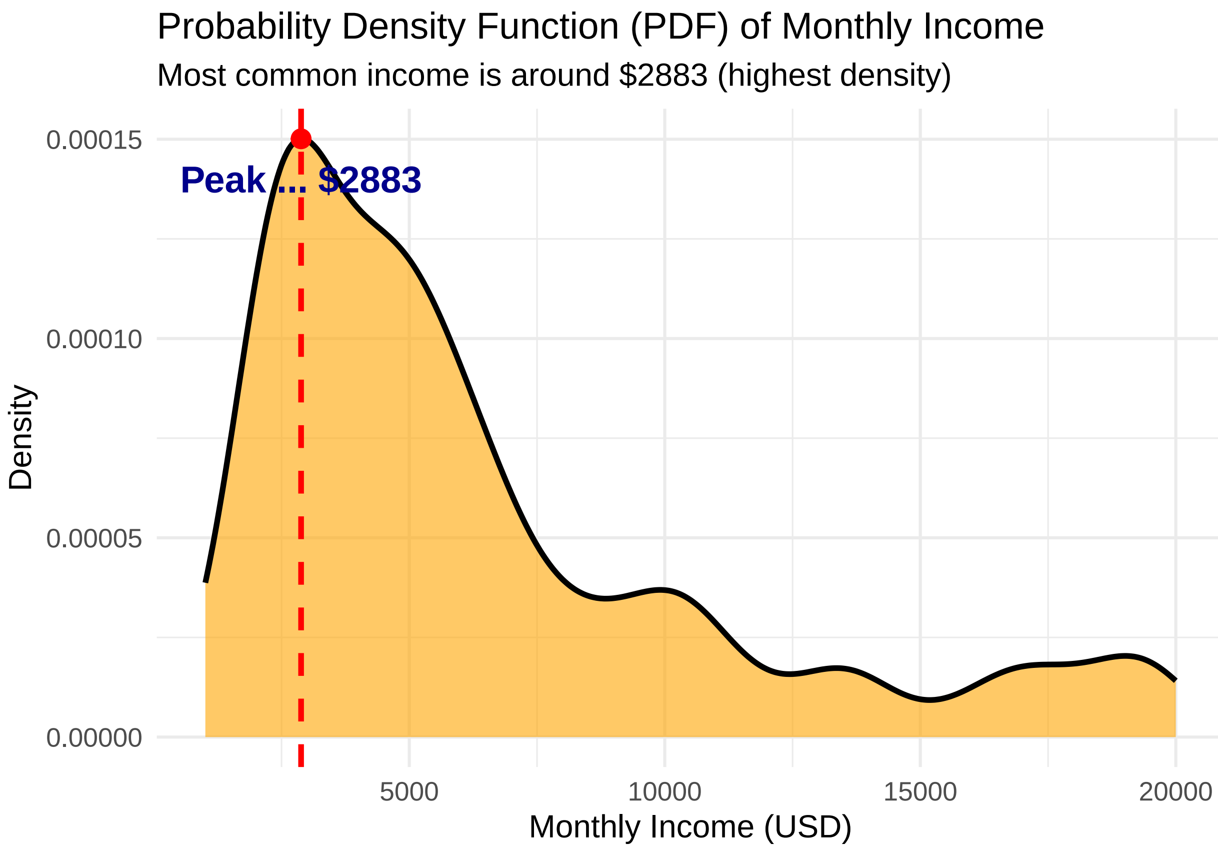
Visualize the Distribution of variable MonthlyIncome

```
# Histogram
ggplot(df_selected, aes(x = MonthlyIncome)) +
  geom_histogram(bins = 30, fill = "skyblue", color = "black") +
  theme_minimal() +
  labs(title = "Distribution of Monthly Income", x = "Monthly Income", y = "Count")
```




```r
# Density Plot

library(ggplot2)

# Density estimation
dens <- density(df_selected$MonthlyIncome)
max_idx <- which.max(dens$y)
max_x <- dens$x[max_idx]
max_y <- dens$y[max_idx]

# Plot
ggplot(df_selected, aes(x = MonthlyIncome)) +
  geom_density(fill = "orange", alpha = 0.6, color = "black", size = 1) +
  geom_vline(xintercept = max_x, linetype = "dashed", color = "red", size = 1) +
  annotate("point", x = max_x, y = max_y, color = "red", size = 3) +
  annotate("text", x = max_x, y = max_y,
           label = paste0("Peak   $", round(max_x,0)),
           vjust = 2, color = "darkblue", fontface = "bold", size = 5) +
  labs(
    title = "Probability Density Function (PDF) of Monthly Income",
    subtitle = paste0("Most common income is around $", round(max_x,0),
                      " (highest density)"),
    x = "Monthly Income (USD)",
    y = "Density"
  ) +
  theme_minimal(base_size = 12)
```

```
## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Peak  $2883' in 'mbcsToSbcs': dot substituted for <e2>

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Peak  $2883' in 'mbcsToSbcs': dot substituted for <89>

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Peak  $2883' in 'mbcsToSbcs': dot substituted for <88>
```



Calculate Expectation (Mean) and Variance

```r
# Mean (expectation)
mean_income <- mean(df_selected$MonthlyIncome)
mean_income
```

```
## [1] 6502.931
```

```r
# Variance
var_income <- var(df_selected$MonthlyIncome)
var_income
```

```
## [1] 22164857
```

```r
# Standard deviation
sd_income <- sd(df_selected$MonthlyIncome)
sd_income
```

```
## [1] 4707.957
```

Empirical Cumulative Distribution Function (CDF)

```r
# Empirical CDF
cdf_income <- ecdf(df_selected$MonthlyIncome)

# Plot the CDF
# plot(cdf_income, main = "Empirical CDF of Monthly Income", xlab = "Monthly Income", ylab = "F(x)")

# Example: Proportion of employees earning <= $5000
cdf_income(5000)
```

```
## [1] 0.5095238
```

```r
# Empirical CDF
cdf_income <- ecdf(df_selected$MonthlyIncome)

p_le_5000 <- cdf_income(5000)                 # P(X   5000)
median_inc <- median(df_selected$MonthlyIncome)

# Plot
plot(cdf_income,
     main = "Empirical CDF of Monthly Income",
     xlab = "Monthly Income (USD)", ylab = "F(x)",
     col = "black", lwd = 2, pch = 16, cex = 0.8)


# Show P(X   5000)
abline(v = 5000,           col = "red",  lty = 2)
abline(h = p_le_5000,      col = "red",  lty = 2)
points(5000, p_le_5000,    col = "red",  pch = 19, cex = 1.3)
text(5000, p_le_5000,
     labels = paste0(" P(X   5000) = ", round(p_le_5000, 3)),
     pos = 4, col = "red", cex = 0.95, font = 2)
```

```
## Warning in text.default(5000, p_le_5000, labels = paste0(" P(X   5000) = ", :
## conversion failure on ' P(X   5000) = 0.51' in 'mbcsToSbcs': dot substituted
## for <e2>

## Warning in text.default(5000, p_le_5000, labels = paste0(" P(X   5000) = ", :
## conversion failure on ' P(X   5000) = 0.51' in 'mbcsToSbcs': dot substituted
## for <89>

## Warning in text.default(5000, p_le_5000, labels = paste0(" P(X   5000) = ", :
## conversion failure on ' P(X   5000) = 0.51' in 'mbcsToSbcs': dot substituted
## for <a4>
```

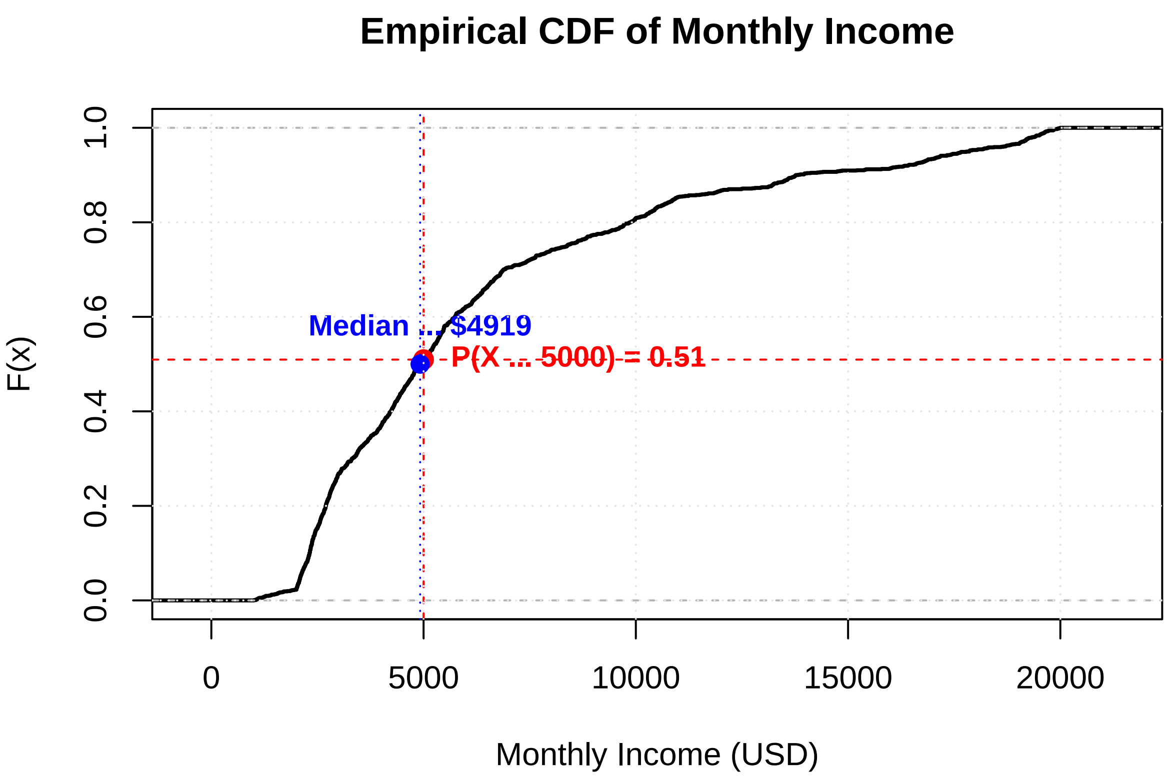
```r
abline(v = median_inc, col = "blue", lty = 3)
points(median_inc, 0.5, col = "blue", pch = 19, cex = 1.2) #
text(median_inc, 0.52,
     labels = paste0("Median   $", round(median_inc, 0)),
     pos = 3, col = "blue", cex = 0.9, font = 2)
```

```
## Warning in text.default(median_inc, 0.52, labels = paste0("Median   $", :
## conversion failure on 'Median   $4919' in 'mbcsToSbcs': dot substituted for
## <e2>

## Warning in text.default(median_inc, 0.52, labels = paste0("Median   $", :
```

```
## conversion failure on 'Median   $4919' in 'mbcsToSbcs': dot substituted for
## <89>
```

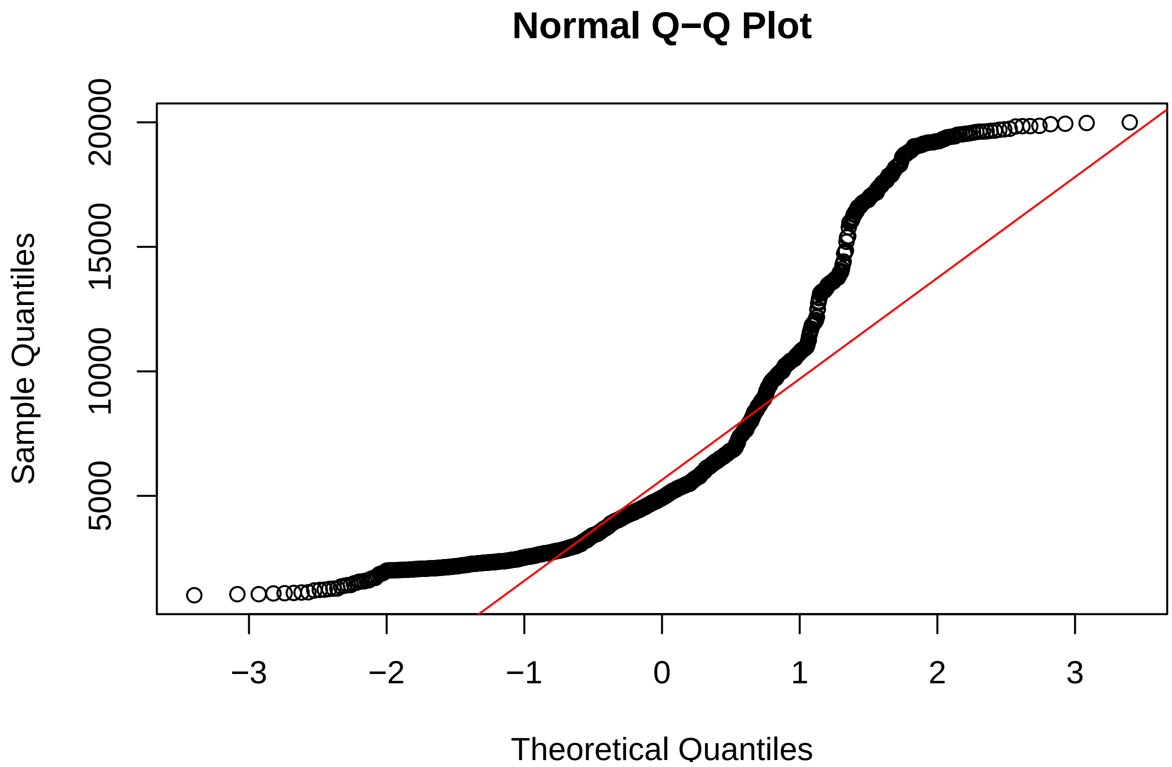
```
## Warning in text.default(median_inc, 0.52, labels = paste0("Median   $", :
## conversion failure on 'Median   $4919' in 'mbcsToSbcs': dot substituted for
## <88>
```

```r
grid(col = "gray90")
```



Test for Normality

```r
# QQ Plot
qqnorm(df_selected$MonthlyIncome)
qqline(df_selected$MonthlyIncome, col = "red")
```


Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
5000
10000
15000
20000
−3
−2
−1
0
1
2
3